Time Series Analysis

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BOOKS

- Text Book

(1) Shumway, R. H., & Stoffer, D. S. (2017). Time series analysis and its applications: with R examples. Springer.

(2) Brockwell, P. J., & Davis, R. A. (2016). Introduction to time series and forecasting. springer.

- Reference book

(1) Cryer, J. D., & Kellet, N. (1991). Time series analysis. Royal Victorian Institute for the Blind. Tertiary Resource Service.

(2) Box, G. E., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2015). Time series analysis: forecasting and control. John Wiley & Sons.

Introduction

Time series

Definition

Time series is a collection of random variables $\{X_t | \ t \in T\}$ over a time index set T, which might be a finite, countably infinite or uncountable set.

- Realized values: What we observe are the realized values of the time series i.e. the data set is $\{X_1 = x_1, X_2 = x_2, \cdots X_n = x_n\}$, where the x_is are some numeric or categorical values.

Categories of Time series

- Discerte time series: If T is a countable set then it is a discerte time series.
- Continuous time series: If T is an interval then it is a continuous time series.
- Note: Discerte or continuous are the adjectives of time but NOT of random variable X_t

Operators

- Back-shift operator B, such that $B^h X_t = X_{t-h}$
- Difference operator: $I - B = \nabla$ which gives

$$\nabla X_t = X_t - X_{t-1} = (I - B)X_t$$

which implies

$$\nabla^h X_t = (I - B)^h X_t$$

- Seasonal difference

$$\nabla_s X_t = (1 - B^s)X_t$$

Polynomial $y = 1 + 2x + 3x^2$

```
x<-seq(1,1.5, by=0.1)
y<-1+2*x+3*x^2
l<-length(x)
dtable<-array(0,dim = c(l,l+2))
dtable[,1]<-x
dtable[,2]<-y
for(i in 1: 3){
  dtable[1:(l-i),(i+2)]<-diff(y,1,i)
}
round(dtable, 2)
```

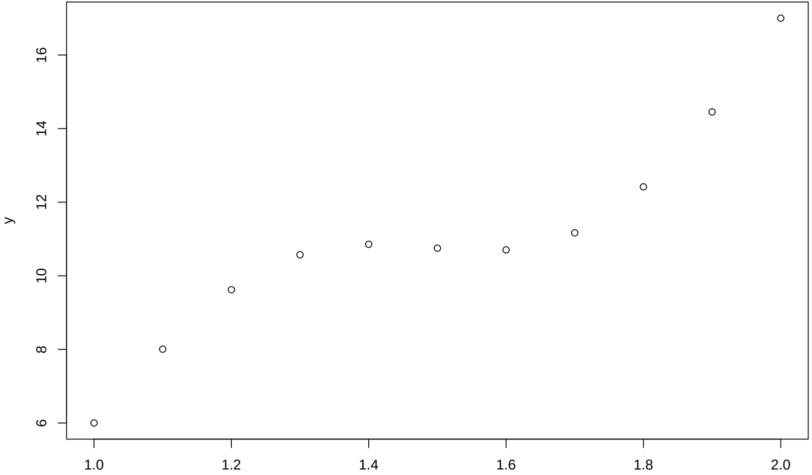
Difference Table

Polynomial $y = 1 + 2x + 3x^2$

```
##       [,1]  [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]   1.0  6.00 0.83 0.06    0    0    0    0
## [2,]   1.1  6.83 0.89 0.06    0    0    0    0
## [3,]   1.2  7.72 0.95 0.06    0    0    0    0
## [4,]   1.3  8.67 1.01 0.06    0    0    0    0
## [5,]   1.4  9.68 1.07 0.00    0    0    0    0
## [6,]   1.5 10.75 0.00 0.00    0    0    0    0
```

Polynomial & periodic $y = 1 + 2x + 3x^2 + \sin(2\pi x)$

```
x<-seq(1,2, by=0.1)
y<-1+2*x+3*x^2+2*sin(2*pi*x)
plot(y~x)
```



Difference Table

Polynomial & periodic $y = 1 + 2x + 3x^2 + \sin(2\pi x)$

```
##         [,1]  [,2]  [,3]  [,4]  [,5]  [,6]  [,7] [,8]  [,9] [,10]
##  [1,]  6.00  2.01 -0.39 -0.28  0.28  0.00 -0.11 0.04  0.03 -0.03
##  [2,]  8.01  1.62 -0.67  0.00  0.28 -0.11 -0.07 0.07  0.00 -0.03
##  [3,]  9.62  0.95 -0.67  0.28  0.17 -0.17  0.00 0.07 -0.03  0.00
##  [4,] 10.57  0.28 -0.39  0.45  0.00 -0.17  0.07 0.04  0.00  0.00
##  [5,] 10.86 -0.11  0.06  0.45 -0.17 -0.11  0.11 0.00  0.00  0.00
##  [6,] 10.75 -0.05  0.51  0.28 -0.28  0.00  0.00 0.00  0.00  0.00
##  [7,] 10.70  0.46  0.79  0.00 -0.28  0.00  0.00 0.00  0.00  0.00
##  [8,] 11.17  1.25  0.79 -0.28  0.00  0.00  0.00 0.00  0.00  0.00
##  [9,] 12.42  2.04  0.51  0.00  0.00  0.00  0.00 0.00  0.00  0.00
## [10,] 14.45  2.55  0.00  0.00  0.00  0.00  0.00 0.00  0.00  0.00
## [11,] 17.00  0.00  0.00  0.00  0.00  0.00  0.00 0.00  0.00  0.00
```

Difference Table

Polynomial & periodic & random error $y = 1 + 2x + 3x^2 + \sin(2\pi x) + \epsilon$

```
x<-seq(1,2, by=0.1)
l<-length(x)
y<-1+2*x+3*x^2+ 2*sin(2*pi*x)+ rnorm(l,0,0.1)

dtable<-array(0,dim = c(l,l+2))
dtable[,1]<-x
dtable[,2]<-y
for(i in 1: (l-1)){
  dtable[1:(l-i),(i+2)]<-diff(y,1,i)
}
round(dtable[,2:11], 2)
```

Difference Table

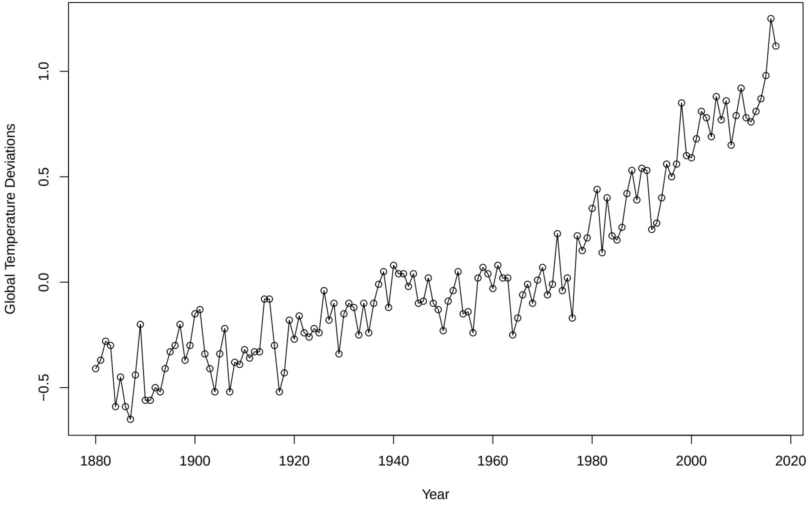
Polynomial & periodic & random error $y = 1 + 2x + 3x^2 + \sin(2\pi x) + \epsilon$

```
##          [,1]  [,2]  [,3]  [,4]  [,5]  [,6]  [,7]  [,8]  [,9] [,10
##  [1,]  6.04  1.91 -0.21 -0.72  1.38 -2.23  3.64 -5.33  6.72 -7.1
##  [2,]  7.95  1.69 -0.94  0.65 -0.85  1.41 -1.69  1.38 -0.40 -1.5
##  [3,]  9.64  0.75 -0.28 -0.20  0.56 -0.28 -0.31  0.99 -1.93  0.0
##  [4,] 10.39  0.47 -0.48  0.36  0.27 -0.59  0.68 -0.94  0.00  0.0
##  [5,] 10.86 -0.02 -0.13  0.63 -0.32  0.09 -0.26  0.00  0.00  0.0
##  [6,] 10.85 -0.14  0.50  0.32 -0.23 -0.17  0.00  0.00  0.00  0.0
##  [7,] 10.70  0.36  0.82  0.09 -0.40  0.00  0.00  0.00  0.00  0.0
##  [8,] 11.07  1.18  0.91 -0.31  0.00  0.00  0.00  0.00  0.00  0.0
##  [9,] 12.25  2.09  0.60  0.00  0.00  0.00  0.00  0.00  0.00  0.0
## [10,] 14.34  2.70  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.0
## [11,] 17.04  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.0
```

Meteorological station data were used to estimate the global annual-mean surface air temperature deviation from 1880 to 2018.
source : https://data.giss.nasa.gov/gistemp/graphs/

```
gtmp<-read.csv("graph.csv",header = T,sep =",")
plot(gtmp$No_Smoothing~gtmp$Year, type="o",
     ylab="Global Temperature Deviations", xlab='Year')
```

Global Temperature Deviations



White Noise

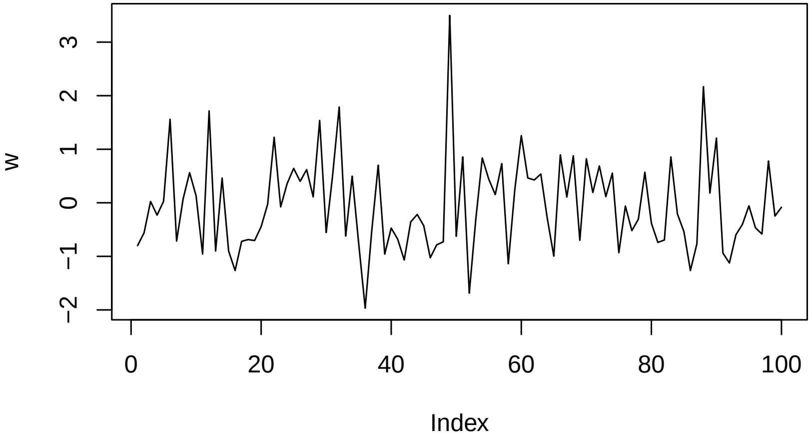
Definition

The time series generated from uncorrelated variables with zero mean and fixed finite variance is called white noise [$Notation \ \ W_t \sim WN(0, \sigma_w^2)$]

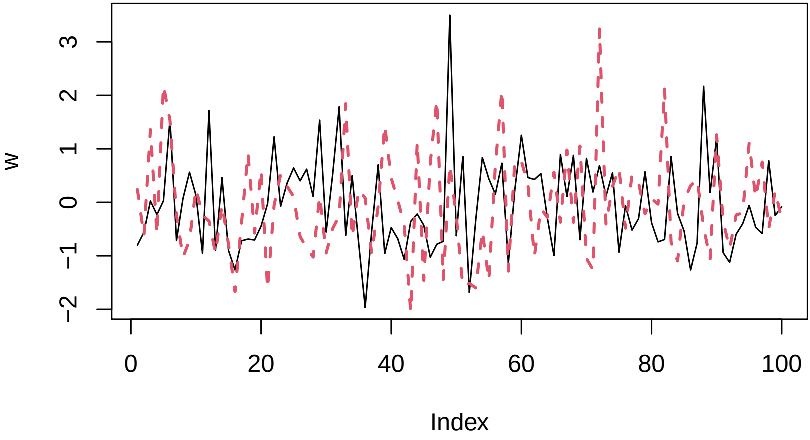
- Example1: $X_t = W_t = \begin{cases} N(0,1) \text{ if t is even }, \\ Exp(1) - 1 \text{ if t is odd }. \end{cases}$
- Example2 : $X_t = W_t \ \ i.i.d. \ \ N(0, \sigma^2)$

White Noise (Example)

```
set.seed(123); n<-100; w<-array(0,dim = c(n))
e<-seq(from = 2,to = n, by=2) ; o<-(e-1);
w[e]<-rnorm((n/2),0,1) ; w[o]<-rexp((n/2),1)-1;
plot(w, type = "l")
```



```
set.seed(123); n<-100; w<-array(0,dim = c(n))
e<-seq(from = 2,to = n, by=2) ; o<-(e-1);
w[e]<-rnorm((n/2),0,1) ; w[o]<-rexp((n/2),1)-1;
plot(w, type = "l"); lines(rnorm(n),col=2,lty=2,lwd=2)
```



Trend

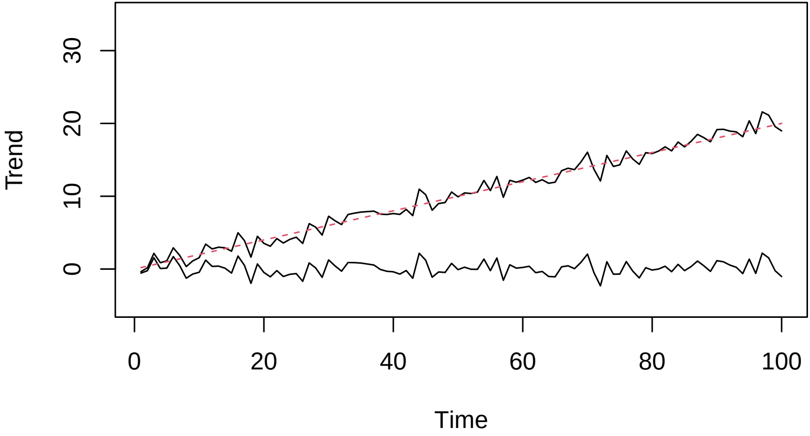
Definition

A deterministic pattern (T_t) which persist through out in the time sires. For example $X_t = T_t + W_t$

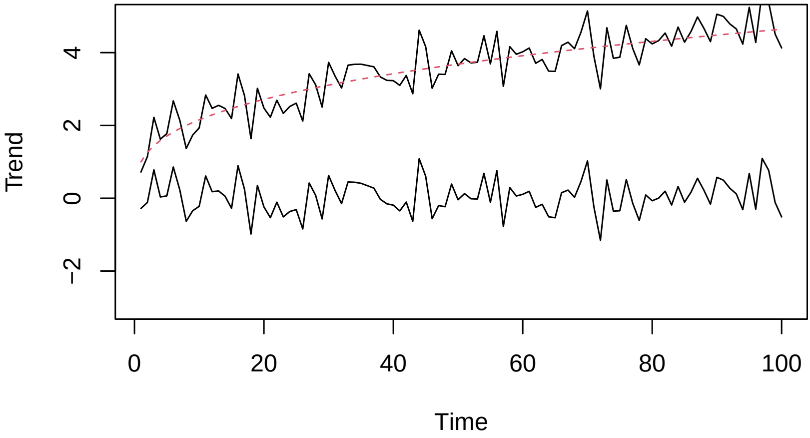
- Example: $X_t = T_t + W_t = 0.2t + W_t$
- Example: $X_t = T_t + W_t = t^{(1/3)} + W_t$

Trend (Example)

```
set.seed(123); n<-100;  w <- rnorm(n,0,1);
xw<-w; xz<-0.2*(1:n)+w
plot.ts(xw, ylim=c(-5,35), ylab="Trend")
lines(xz); lines(0.2*(1:n), col=2, lty="dashed")
```



```
set.seed(123); n<-100;  w <- rnorm(n,0,.5);
xw<-w; xz<-(1:n)^(1/3)+w
plot.ts(xw, ylim=c(-3,5), ylab="Trend")
lines(xz); lines((1:n)^(1/3), col=2, lty="dashed")
```



Seasonality

Definition

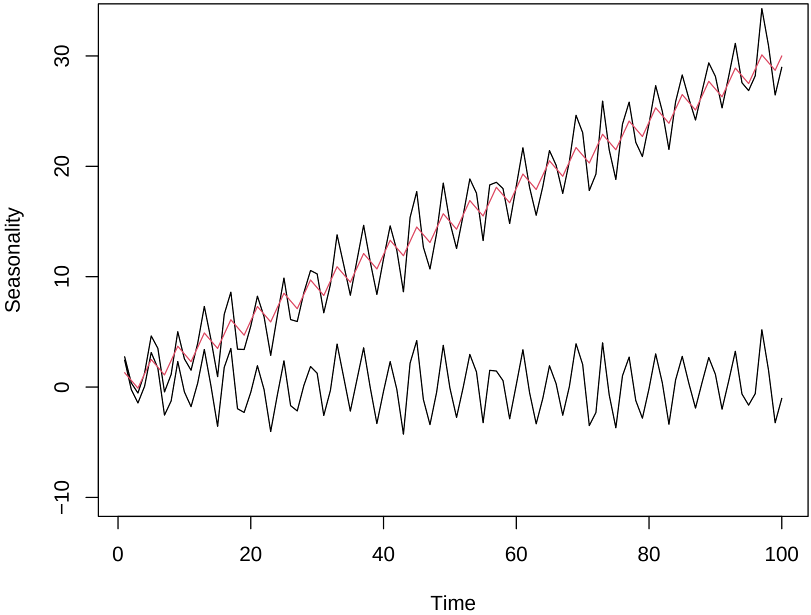
A deterministic pattern (S_t) which returns in the time sires after a fixed interval. For example $X_t = T_t + S_t + W_t$

- Example: $X_t = T_t + S_t + W_t = 0.3t + 3\sin(4.5t\pi) + W_t$
- Example: $X_t = S_t + W_t = 3\sin(4.5t\pi) + W_t$

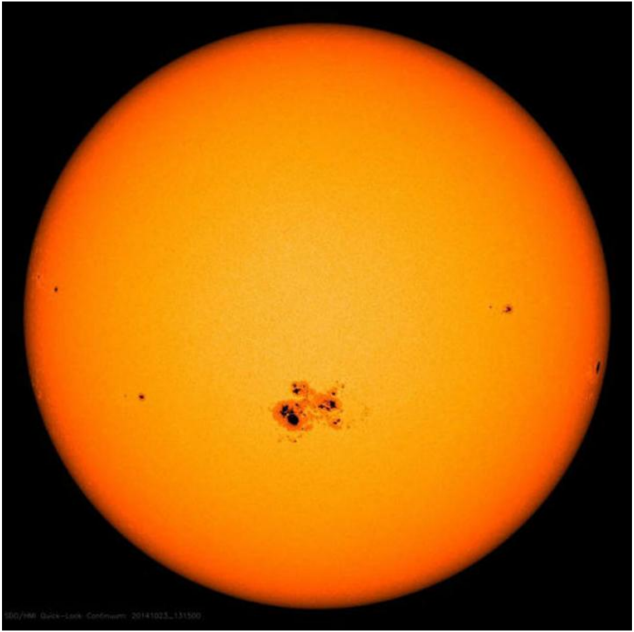
Seasonality (Example)

```
set.seed(123); n<-100;
w <- rnorm(n,0,1);
xw<-0.3*(1:n) + 3*sin(4.5*(1:n)*pi)+w;
xz<-  3*sin(4.5*(1:n)*pi)+w;
plot.ts(xw, ylim=c(-10,33), ylab="Seasonality");
lines(xz);
lines(0.3*(1:n)+ sin(4.5*(1:n)*pi), col=2)
```

Seasonality (Example)



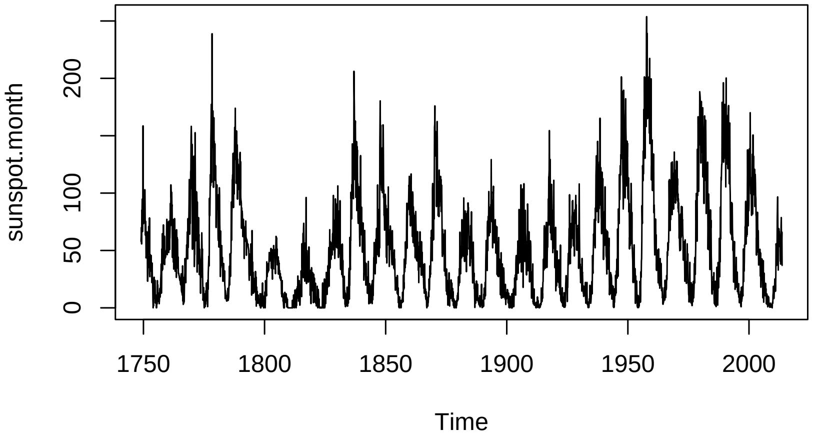
Sunspot



Sunspot data

Monthly numbers of sunspots, as from the World Data Center, aka SIDC. The univariate time series "sunspot.month" contain 2988 observations.

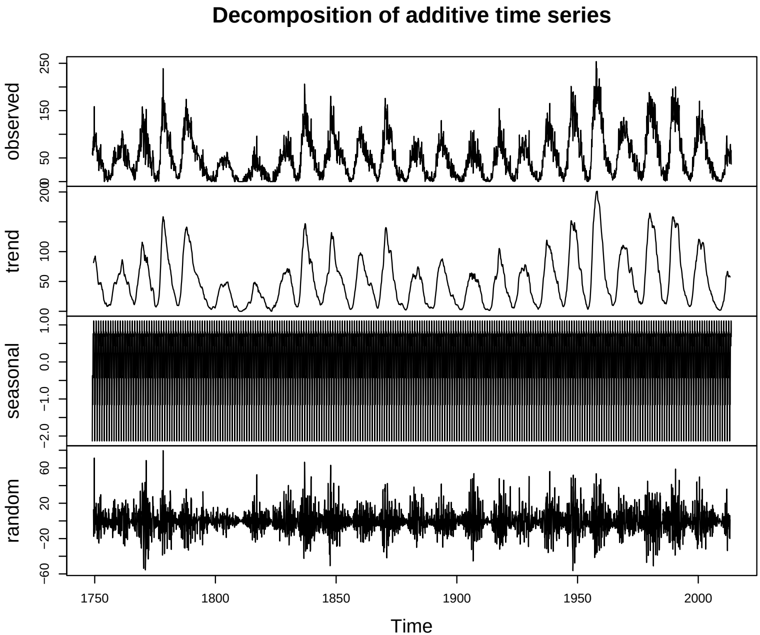
```
plot(sunspot.month)
```



Sunspot data

Additive model : $X_t = T_t + S_t + W_t$

```
plot(decompose(sunspot.month,type = "additive"))
```


Decomposition of additive time series
observed
trend
seasonal
random
Time
250
150
50
0
200
100
50
0
1.00
0.0
−1.0
−2.0
60
20
−20
−60
1750
1800
1850
1900
1950
2000

Stationariry

Mean and Autocovariance

Definition

Suppose that $\{X_t\}$ is a time series with $E|X_t| < \infty$.

Its mean function is

$$\mu_t = E[X_t]$$

.

Definition

Suppose that $\{X_t\}$ is a time series with $E[X_t^2] < \infty$, then its autocovariance function (ACVF) is

$$\gamma_X(s, t) = Cov(X_s, X_t) = E[(X_s - \mu_s)(X_t - \mu_t)]$$

.

Definition

A time series $\{X_t\}$ is said to be weakly stationary if
1. μ_t is independent of t, and
2. For each $h \in \mathbb{Z}$, ACVF $\gamma(t+h,t)$ is independent of t

- Example: $X_t \sim WN(0, \sigma_w^2)$
- NOT an Example: $X_t = \sum_{i=0}^{t} W_i$, where $W \sim WN(0, \sigma_w^2)$

Definition

$\{X_t\}$ is strongly stationary if for all k, h,$t_1, \ldots, t_k$, $x_1, \ldots, x_k$shifting the time axis does not affect the distribution i.e.

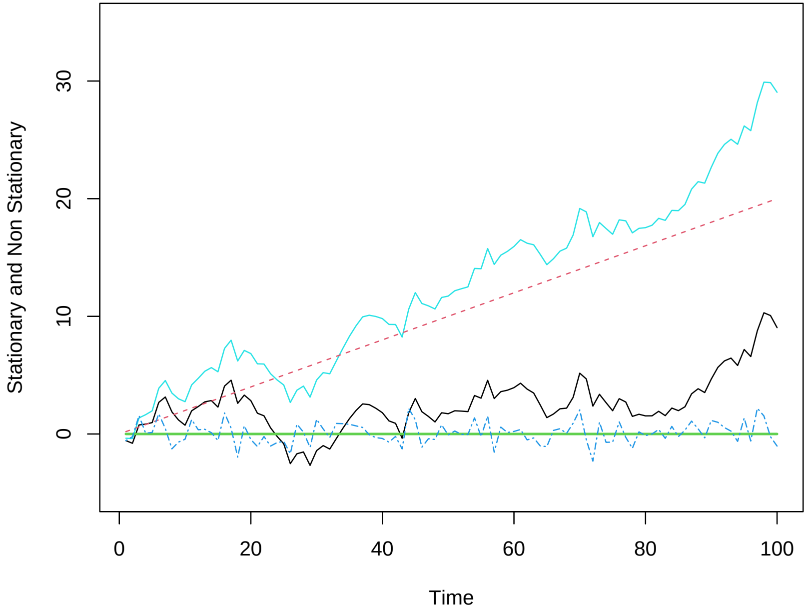
$$P(X_{t_1} \leq x_1, \ldots, X_{t_k} \leq x_k) = P(X_{t_1+h} \leq x_1, ..., X_{t_k+h} \leq x_k)$$

- Example: $X_t \sim WN = N(0, \sigma^2)$
- NOT an Example: Random walk defined as

$$S_t = \sum_{i=0}^{t} X_i, \quad \text{where} \;\; X_i \quad i.i.d. \quad N(0, \sigma^2)$$

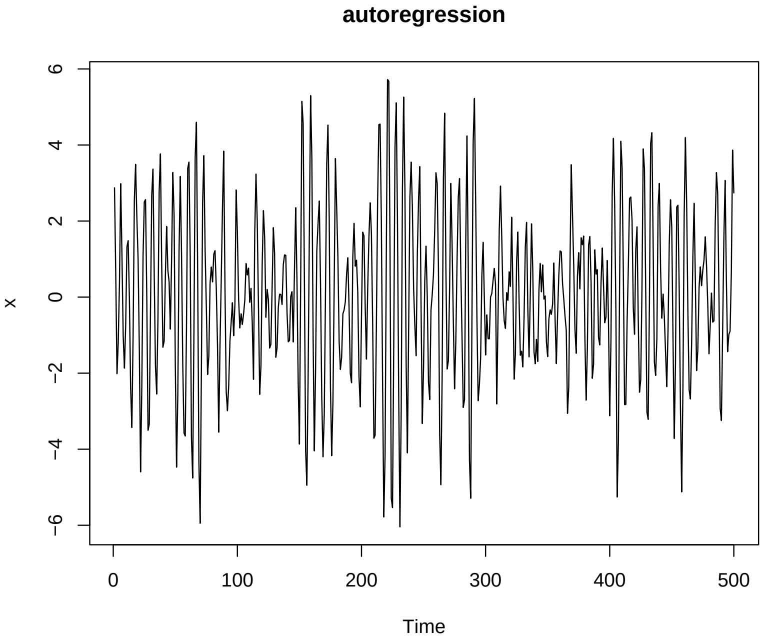
- Strong Stationarity $\implies$ Weak Stationarity

```r
set.seed(123);
n<-100;
w <- rnorm(n,0,1);
z<-w+0.2;
xw<-cumsum(w);
xz<-cumsum(z) ;
plot.ts(xw, ylim=c(-5,35),
        ylab="Stationary and Non Stationary")
lines(xz, col=5);
lines(0.2*(1:n), col=2, lty="dashed");
lines(0.0*(1:n), col=3,lwd=2)
lines(w,lty=6, col=4)
```

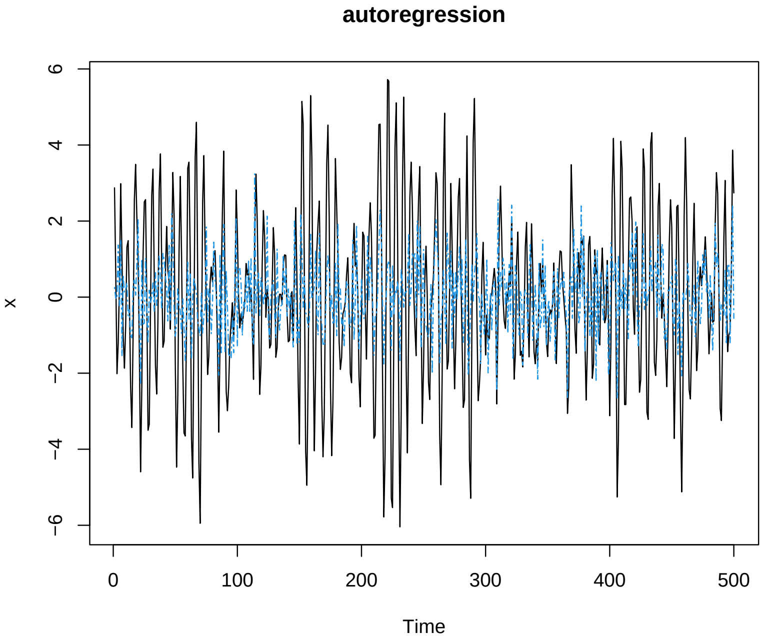

Stationary and Non Stationary
Time
0
10
20
30
0
20
40
60
80
100

Weakly Stationary Time seris (Example)

```
set.seed(123);
n<-550;
w <- rnorm(n,0,1);
x = filter(w, filter=c(1,-.9),
           method="recursive")[-(1:50)]
plot.ts(x, main="autoregression")
lines(w[-(1:50)],lty=6, col=4)
```


autoregression
x
Time
6
4
2
0
−2
−4
−6
0
100
200
300
400
500

Weakly Stationary Time seris (Example)



Properties of a Strongly Stationary Time Series X_t

(1) The random variables X_t are identically distributed.

(2) $(X_t, X_{t+h}) \stackrel{d}{=} (X_1, X_{1+h})$ for all integers t and h.

(3) X_t is weakly stationary if $E(X_t^2) < \infty$ for all t.

(4) Weak stationarity does not imply strongly stationarity.

(5) An i.i.d. sequence is strongly stationary.

Linear process

An important class of weakly stationary time series:

Definition

$$X_t = \mu + \sum_{j=-\infty}^{j=+\infty} \psi_j W_{t-j}$$

where $\mu \in \mathbb{R}$ and $\sum_{j=-\infty}^{j=+\infty} |\psi_j| < \infty$ and $W_j \sim WN(0, \sigma_w^2)$

- $E(X_t) = \mu$
- $\gamma_X(h) = \sigma_w^2 \sum_{j=-\infty}^{j=+\infty} \psi_j \psi_{j-h} < \infty$

WN as a linear proccess

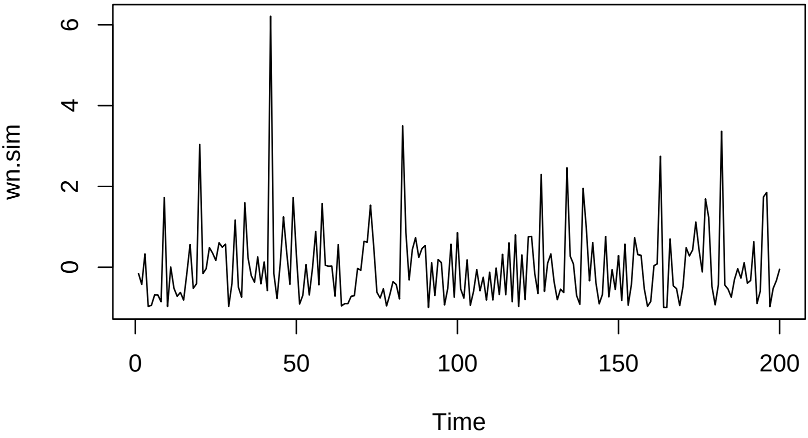
- Time series $X_t = W_t$ where $W_t \sim WN(0, \sigma_w^2)$
- WN process is a linear process with $\mu = 0$ and

$$\psi_j = \begin{cases} 1 & \textit{if} \quad j = 0 \\ 0 & \textit{otherwise} \end{cases}$$

- WN had fixed and finite mean and variance with correlation zero
- WN need not be normally distribute
- WN need not be iid
- WN is always weakly stationary
- Normally distribute WN is strongly stationary
- iid sequence is always white noise and strongly stationary

WN Example

```
set.seed(123); n<-200;
wn.sim<-rexp(n,1)-1
plot.ts(wn.sim)
```



Genaral result

Theorem

Consider a (weakly) stationary time series $\{X_t\}$ with mean zero and define

$$Y_t = \mu + \sum_{j=-\infty}^{j=+\infty} \psi_j X_{t-j}$$

where $\mu \in \mathbb{R}$ *and* $\sum_{j=-\infty}^{j=+\infty} |\psi_j| < \infty$ *then*

$$E(Y_t) = \mu$$

$$\gamma_Y(h) = \sum_{k=-\infty}^{k=+\infty} \sum_{j=-\infty}^{j=+\infty} \psi_k \psi_j \gamma_X(h - j + k) \quad \textit{if exists.}$$

Auto correlation function (ACF)

Suppose that X_t is at least weakly stationary time series with

$$E(X_t) = \mu \quad and \quad \gamma_X(h) = E[(X_t - \mu)(X_{t+h} - \mu)]$$

Definition

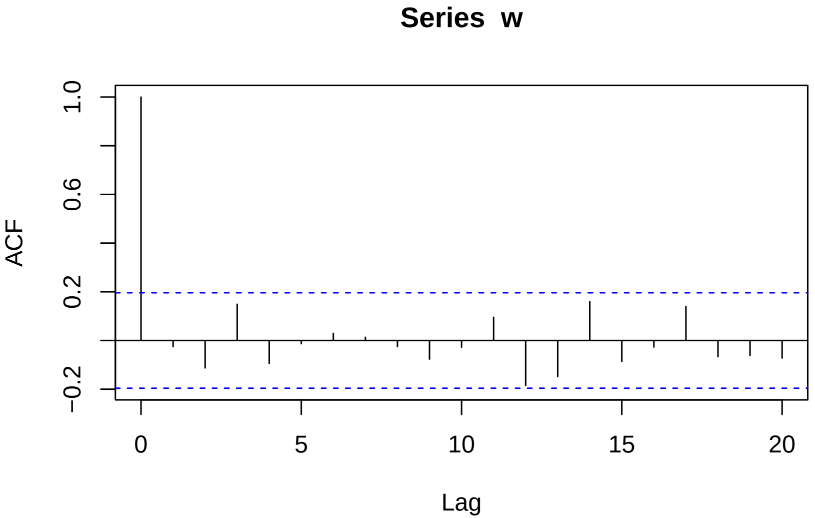
Auto correlation function (ACF) of X_t and X_{t+h} is defined as

$$\rho(h) = \frac{\gamma_X(h)}{\gamma_X(0)}$$

- $\rho(h) = \rho(-h)$
- Let $1 \leq i, j \leq n$ and define a matrix $R = ((\rho(|i-j|)))_{i,j}$ then $\mathbf{a}^T R\mathbf{a} \geq 0$ for all $\mathbf{0} \neq \mathbf{a} \in \mathbb{R}^n$.
- R is positive semidefinite matrix and $\gamma()$ and $\rho()$ are positive semidefinite functions.

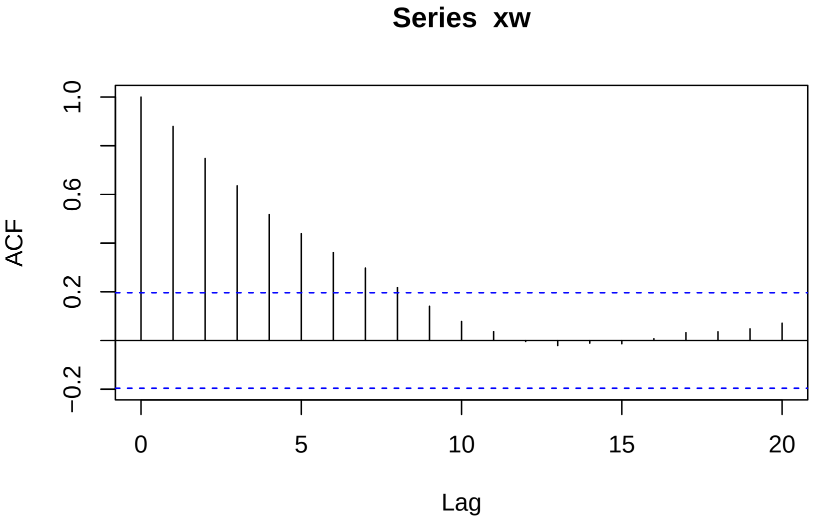
ACF (Example): WN

```
set.seed(123); n<-100;  w <- rnorm(n,0,1); z<-w+0.2;
xw<-cumsum(w); xz<-cumsum(z) ;
acf_xw<-acf(w,type = "correlation",plot = T)
```



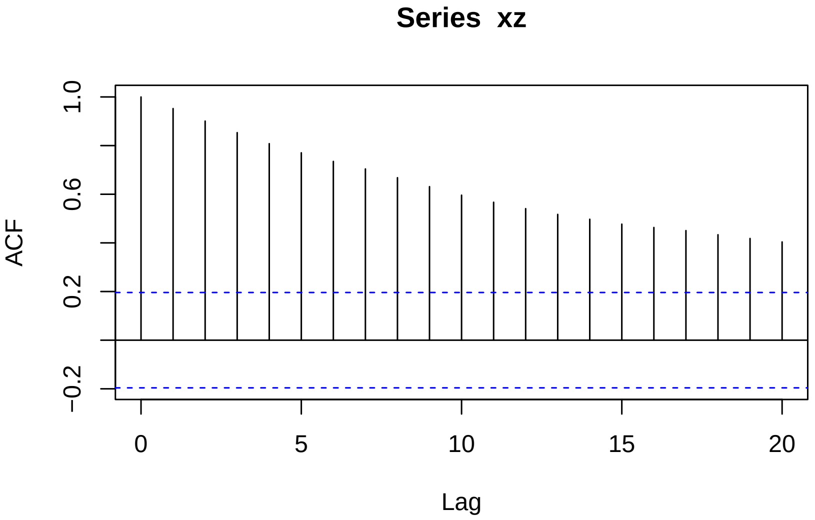
ACF (Example): Random walk

```
set.seed(123); n<-100;  w <- rnorm(n,0,1); z<-w+0.2;
xw<-cumsum(w); xz<-cumsum(z) ;
acf_xw<-acf(xw,type = "correlation",plot = T)
```



ACF (Example): Trend

```
set.seed(123); n<-100;  w <- rnorm(n,0,1); z<-w+0.2;
xw<-cumsum(w); xz<-cumsum(z) ;
acf_xz<-acf(xz,type = "correlation",plot = T)
```



MA(q) processes

Definition

MA(q),a moving average process of order q is defined as

$$X_t = W_t + \theta_1 W_{t-1} + \cdots + \theta_q W_{t-q}$$

Moving average process (MA)

Definition

A time series $\{X_t\}$is said to be an moving average process (of order one) if $X_t = W_t + \theta W_{t-1}$ where $W_t \sim WN(0, \sigma_w^2)$

- MA(1) process is a linear process with $\mu = 0$ and

$$\psi_j = \begin{cases} 1 & \text{if} \quad j = 0 \\ \theta & \text{if} \quad j = 1 \\ 0 & \text{otherwise} \end{cases}$$

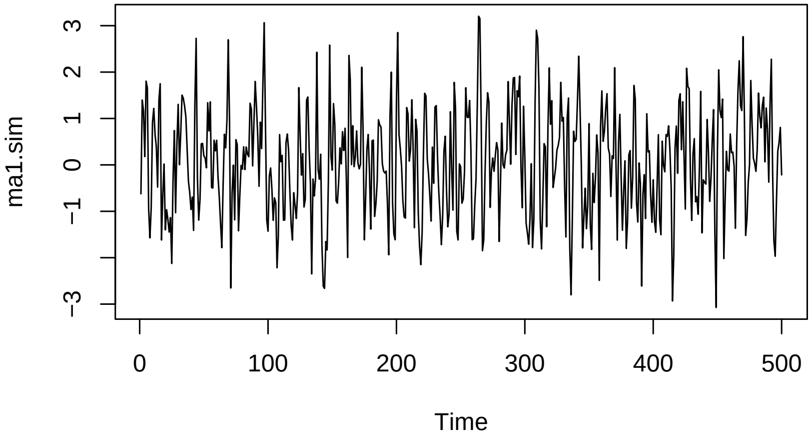
- $E(X_t) = 0$
-

$$\gamma_X(h) = \begin{cases} \sigma_w^2(1+\theta^2) & \text{if} \quad h = 0 \\ \sigma_w^2\theta & \text{if} \quad h = +1, -1 \\ 0 & \text{otherwise} \end{cases}$$

- MA(1) is a at least weakly stationary process

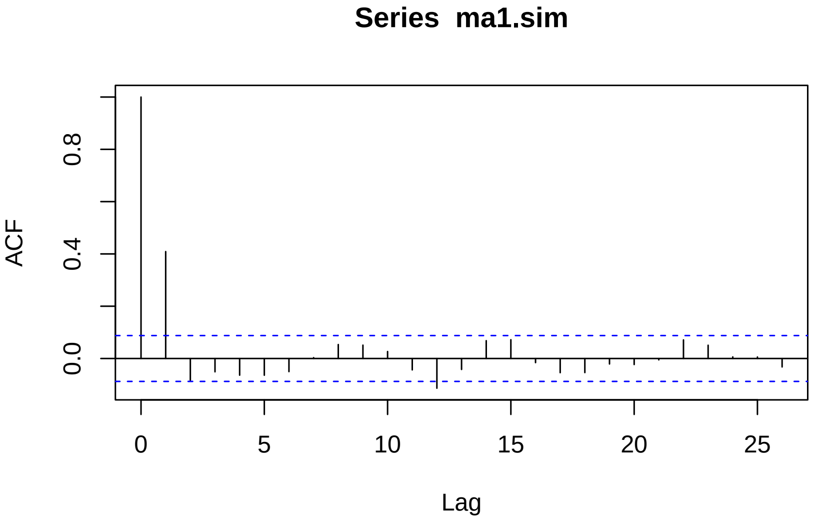
MA(1) Example

```
set.seed(123); n<-500; p<-0; d<-0;q<-1;
ma1.sim<-arima.sim(list(order=c(p,d,q), ma=0.7), n)
ts.plot(ma1.sim)
```



MA(1) Example

```
set.seed(123); n<-500; p<-0; d<-0;q<-1;
ma1.sim<-arima.sim(list(order=c(p,d,q), ma=0.7), n)
acf(ma1.sim,type = "correlation",plot = T)
```



AR(p) processes

Definition

AR(p), an auto-regressive process of order p is defined as

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + W_t$$

Definition

A time series $\{X_t\}$is said to be an auto regressive process (of order one) if $X_t = \phi X_{t-1} + W_t$ where $W_t \sim WN(0, \sigma_w^2)$

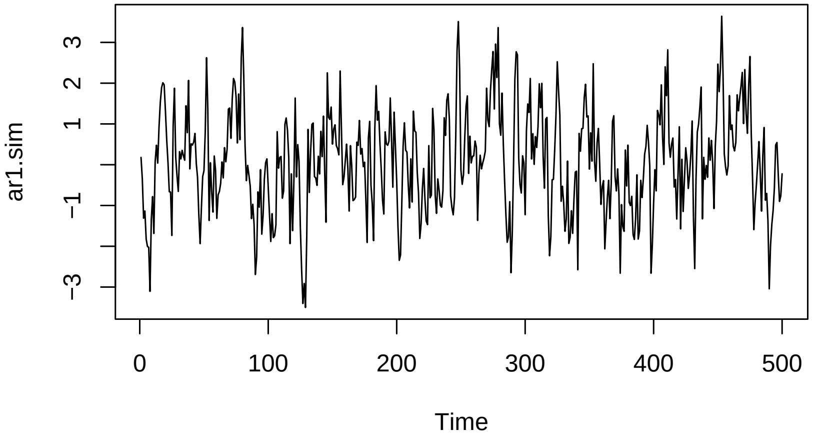
- AR(1) process is a linear process with $\mu = 0$ and

$$\psi_j = \begin{cases} 1 & if \quad j = 0 \\ \phi^j & if \quad j \geq 1 \\ 0 & otherwise \end{cases}$$

- $E(X_t) = 0$
- $\gamma_X(h) = \frac{\sigma_w^2 \phi^{|h|}}{1-\phi^2}$ if $|\phi| < 1$, $\phi \neq 0$ [exponential decay]
- If $\phi = 0$ then AR(1) process is a WN process
- AR(1) is at least a weakly stationary process

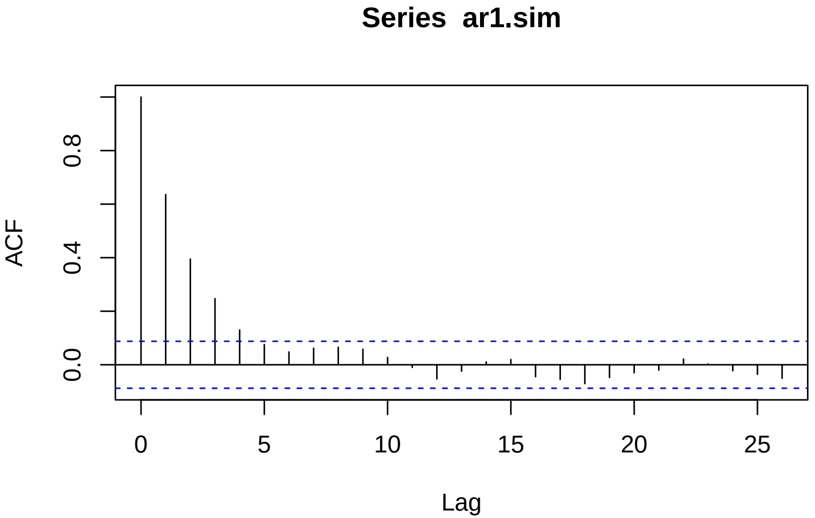
AR(1) Example


```
set.seed(123); n<-500; p<-1; d<-0;q<-0;
ar1.sim<-arima.sim(list(order=c(p,d,q), ar=0.7), n)
ts.plot(ar1.sim)
```



AR(1) Example

```
set.seed(123); n<-500; p<-1; d<-0;q<-0;
ar1.sim<-arima.sim(list(order=c(p,d,q), ar=0.7), n)
acf(ar1.sim,type = "correlation",plot = T)
```



Converges in mean square

Definition

A sequence of random variables $Y_1, Y_2, \cdots$ converges in mean square to Z if for which

$$\lim_{n \to \infty} E(Y_n - Z)^2 = 0$$

- Consider AR(1) process $X_t = \phi X_{t-1} + W_t$ where, $W_t \sim WN(0, \sigma_w^2)$ and $|\phi| < 1$.
- AR(1) is an $MA(\infty)$ process i.e.

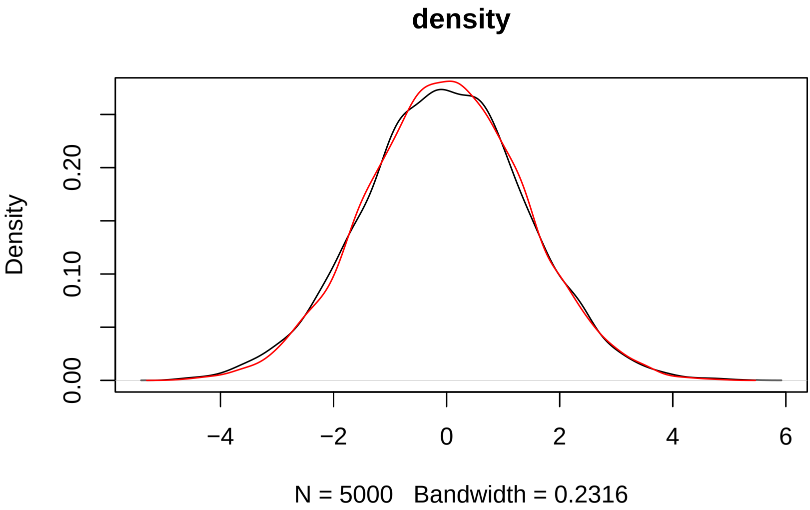
$$\lim_{k \to \infty} E\left(X_t - \sum_{j=0}^{k} \phi^j W_{t-j}\right)^2 = 0$$

- Convergence in mean square $\implies$ Convergence in probability $\implies$ Convergence in distribution, BUT NOT THE OTHER WAY ROUND IN GENERAL.

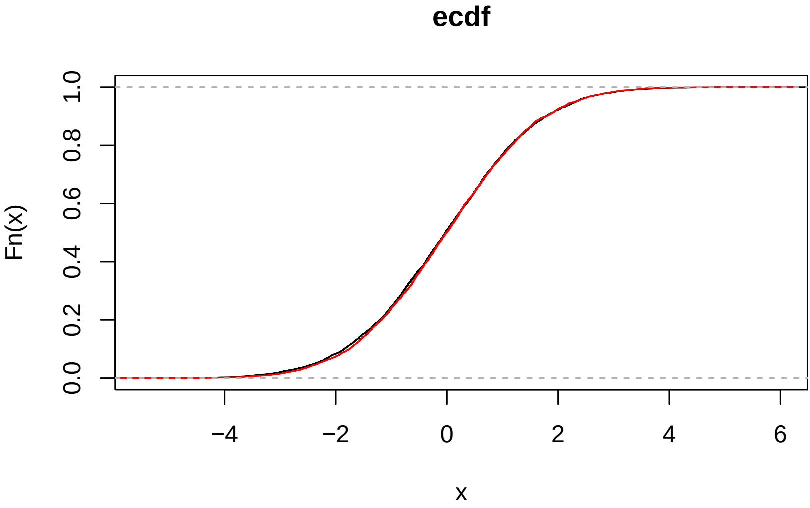
AR(1) and MA(∞) Example

```
set.seed(123);
t<-100
d<-0;
arsim<-numeric(0)
masim<-numeric(0)
for(i in 1 : 5000){
  p<-1; q<-0;
  ar1<-arima.sim(list(order=c(p,d,q), ar=0.7), t)
  p<-0; q<-500;
  mainf<-arima.sim(list(order=c(p,d,q),
                        ma=(0.7)^(seq(1:q))), t)
  arsim[i]<-ar1[t] ;masim[i]<-mainf[t]}
plot(density(arsim), main="density");
lines(density(masim),col="red")
#plot(ecdf(arsim), main="ecdf");
#lines(ecdf(masim),col="red")
```

AR(1) and MA(∞) Example



AR(1) and MA(∞) Example



Use of sampe ACF

Time series	ACF
White Noise	Zero
MA(q)	Zero for $\|h\| > q$
AR(p)	Decays to zero exponentially

Casulity

Definition

A linear process X_t is causal function of W_t if

$$X_t = \left(1 + \sum_{i=1}^{\infty} \phi_i B^i\right) W_t$$

where $\sum_{i=1}^{\infty} |\phi_i| < \infty$

- AR(1) i.e. $X_t = \phi X_{t-1} + W_t$ is casual iff $|\phi| < 1$ i.e. $1 - \phi z$ has a solution out of the unit circle on complex plane $\mathbb{C}$.

Invertibility

Definition

A linear process X_t is invertible function of W_t if there

$$W_t = \left(1 + \sum_{i=1}^{\infty} \theta_i B^i\right) X_t$$

where $\sum_{i=1}^{\infty} |\theta_i| < \infty$

- MA(1) i.e. $X_t = W_t + \theta W_{t-1}$ is invertable iff $|\theta| < 1$ i.e. $1 + \theta z$ has a solution out of the unit circle on complex plane $\mathbb{C}$.

Auto regressive-Moving average process (ARMA (p,q))

Definition

An **ARMA(p,q)** process X_t is a stationary process that satisfies

$$X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p} = W_t + \theta_1 W_{t-1} + \cdots + \theta_q W_{t-q}$$

where $W_i \sim WN(0, \sigma_w^2)$.

- AR(p)= ARMA(p,0)
- MA(q)= ARMA(0,q)

$ARMA(p,q)$

- Back-shift operator B, such that $BX_t = X_{t-1}$
- $\Phi_p(z) = 1 - \sum_{i=1}^{p} \phi_i z^i$ if $\phi_p \neq 0$
- $\Theta_q(z) = 1 + \sum_{i=1}^{q} \theta_i z^i$ if $\theta_q \neq 0$
- Then ARMA(p,q) model can be written as

$$\Phi_p(B)X_t = \Theta_q(B)W_t$$

if $\Phi_p(z)$ and $\Theta_q(z)$ have no common factor i.e. not a lower order ARMA model is possible.

$$X_t - \phi X_{t-1} = W_t + \theta W_{t-1} \quad with \quad |\phi| < 1$$

- $E(X_t) = 0$
-

$$\gamma(h) = \begin{cases} \sigma^2 \left[1 + \frac{(\theta+\phi)^2}{1-\phi^2}\right] & if \quad h = 0 \\ \sigma^2 \left[\theta + \phi + \frac{\phi(\theta+\phi)^2}{1-\phi^2}\right] & if \quad h = 1 \\ \gamma(1)\phi^{h-1} & if \quad h \geq 2 \end{cases}$$

ARMA(p,q): Stationarity, causality, and invertibility

Theorem

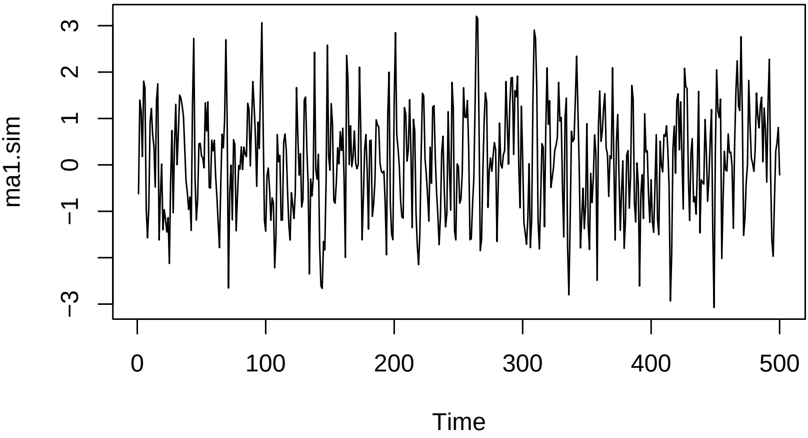
If Φ_p and Θ_q have no common factors, a (unique) stationary solution to $\Phi_p(B)X_t = \Theta_q(B)W_t$ exists iff the roots of $\Phi_p(z)$ avoid the unit circle i.e.

$$|z| = 1 \implies \Phi_p(z) \neq 0.$$

- This ARMA(p,q) process is causal iff the roots of $\Phi_p(z)$ are out- side the unit circle i.e. $|z| \leq 1 \implies \Phi_p(z) \neq 0$.
- It is invertible iff the roots of $\Theta_q(z)$ are outside the unit circle i.e. $|z| \leq 1 \implies \Theta_q(z) \neq 0$.

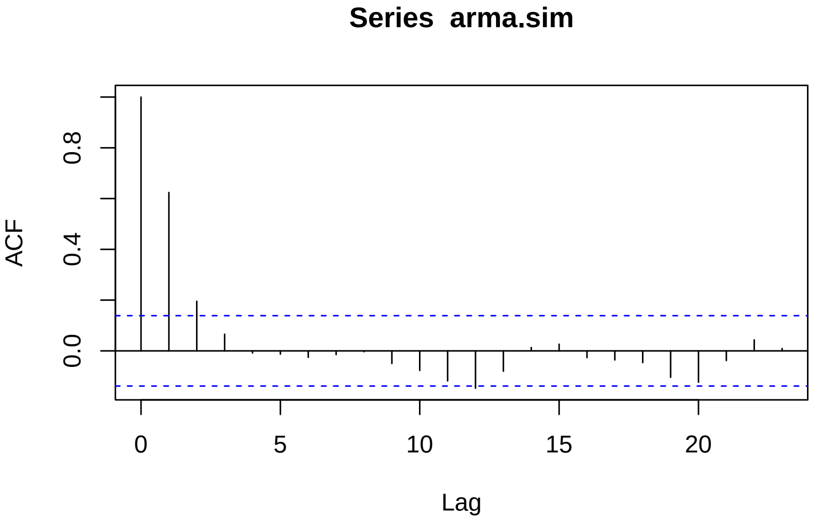
ARMA(1,1) Example

```
set.seed(123); n<-200; p<-1; d<-0;q<-1;
arma.sim<-arima.sim(list(order=c(p,d,q), ma=0.7, ar=.4), n)
ts.plot(ma1.sim)
```



ARMA(1,1) Example

```
set.seed(123); n<-200; p<-1; d<-0;q<-1;
arma.sim<-arima.sim(list(order=c(p,d,q), ma=0.7,ar=c(0.4)), n)
acf(arma.sim,type = "correlation",plot = T)
```



Partial Auto correlation function (PACF)

Definition

Partial Auto correlation function $\alpha(h)$ between X_h and X_0 for given $X_1, \cdots, X_{h-1}$ the correlation between the linear prediction errors $X_h - Ln(X_h|X_1, \dots, X_{h-1})$ and $X_0 - Ln(X_0|X_1, \dots, X_{h-1})$

$$\alpha(h) = \frac{\gamma(h) - \widetilde{\boldsymbol{\gamma}}_{h-1}^{T}(1)\Gamma_{h-1}^{-1}\boldsymbol{\gamma}_{h-1}(1)}{\gamma(0) - \boldsymbol{\gamma}_{h-1}^{T}(1)\Gamma_{h-1}^{-1}\boldsymbol{\gamma}_{h-1}(1)}$$

where

$$\widetilde{\boldsymbol{\gamma}}_{h-1}^{T}(1) = (\gamma(h-1)^{T}, \dots, \gamma(1))$$

and

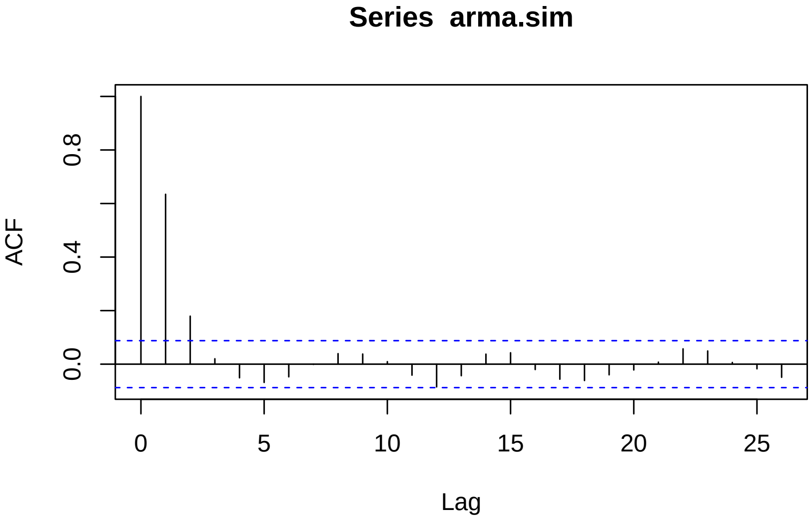
$$\boldsymbol{\gamma}_{h-1}^{T}(1) = (\gamma(1), \dots, \gamma(h-1))^{T}$$

and

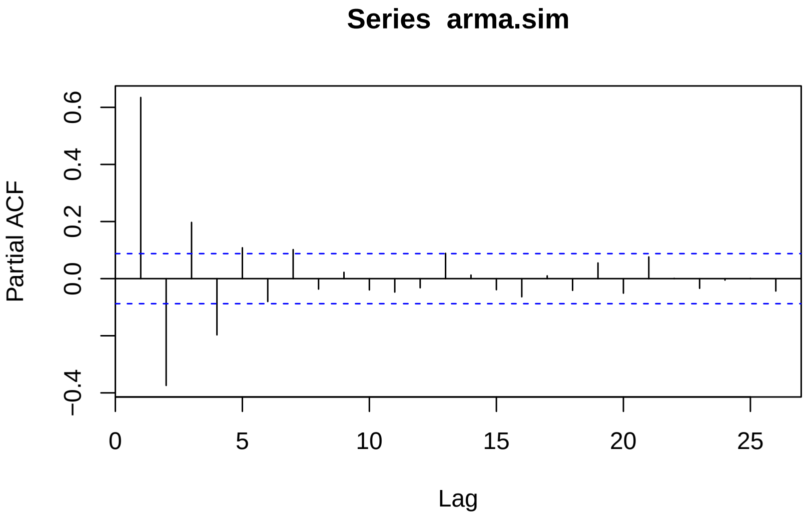
$$\Gamma_{h-1} = ((\gamma(|i-j|)))_{ij}$$

Classification

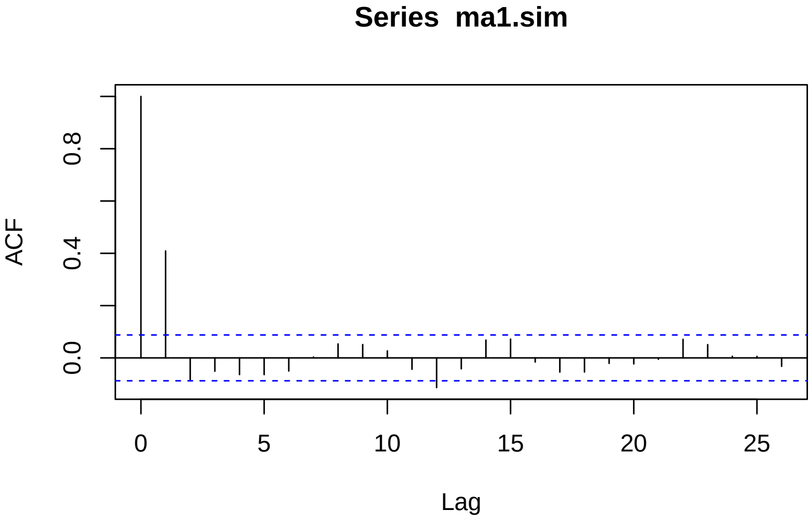
Model:	ACF:	PACF:
AR(p)	decays	zero for $h > p$
MA(q)	zero for $h > q$	decays
ARMA(p,q)	decays	decays

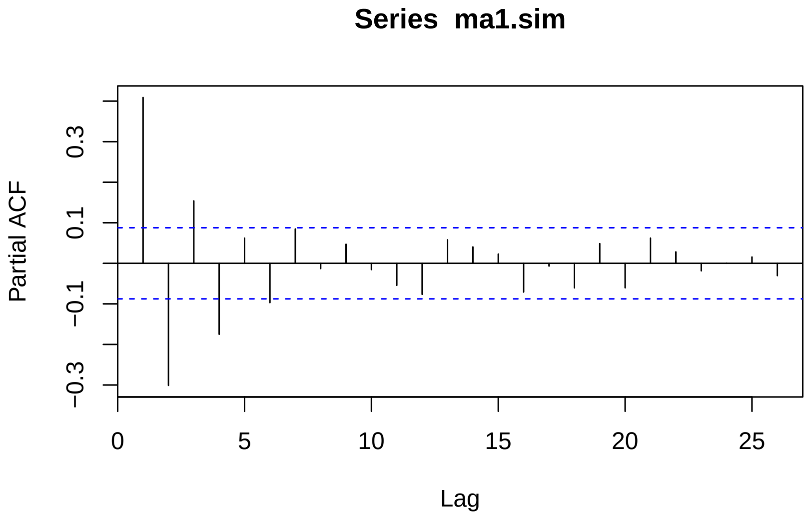

Series arma.sim
ACF
0.0
0.4
0.8
Lag
0
5
10
15
20
25

ARMA(1,1) Example

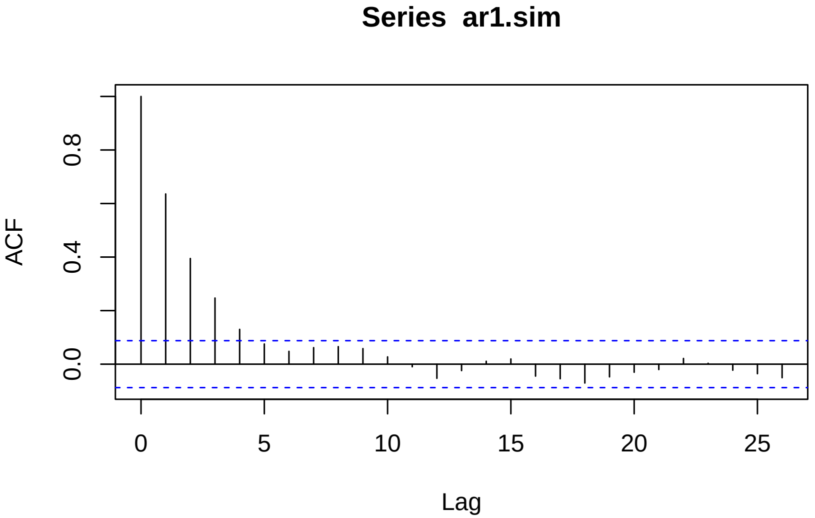


MA(1) Example

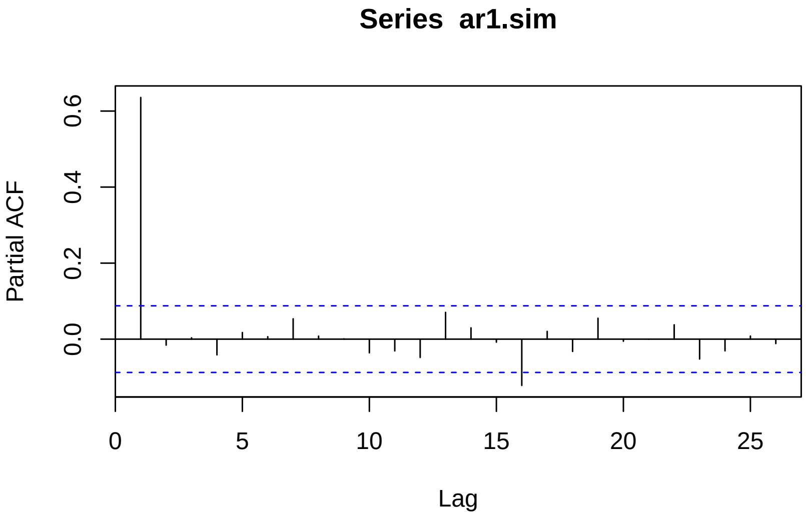



Series ma1.sim
Partial ACF
0.3
0.1
−0.1
−0.3
0
5
10
15
20
25
Lag

AR(1) Example



AR(1) Example



Esimation

Sample MEAN, ACVF & ACF

For observations $x_1, \ldots, x_n$ of a time series,

$$Sample \quad mean \quad \bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

The sample autocovariance function is

$$\hat{\gamma}(h) = \frac{1}{n} \sum_{t=1}^{n-|h|} (x_t - \bar{x})(x_{t+|h|} - \bar{x}) \quad if \quad -n < h < n$$

The sample auto-correlation function is

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}$$

About sample mean $\bar{x}$

- $E(\bar{x}) = \mu$ [unbiased estimator]
- $Var(\bar{x}) = \frac{1}{n}\sum_{h=-n}^{n}\left(1 - \frac{|h|}{n}\right)\gamma(h) \to 0 \quad as \quad n \to \infty$
- $\widehat{Var(\bar{x})} = \frac{1}{n}\sum_{h=-n}^{n}\left(1 - \frac{|h|}{n}\right)\widehat{\gamma(h)}$

Definition

Long run variance

$$\lim_{n\to\infty} nVar(\bar{x}) = \lim_{n\to\infty}\sum_{h=-\infty}^{\infty}\left(1 - \frac{|h|}{n}\right)\gamma(h) = \sigma_w^2\left(\sum_{j=-\infty}^{\infty}\psi_j\right)^2$$

- $nVar(\bar{x}) \to \sum_{-\infty}^{\infty}\gamma(h)$ if $\sum_{-\infty}^{\infty}|\gamma(h)| < \infty$ as $n \uparrow \infty$.

About sample mean $\bar{x}$

If $\{X_t\}$ is a stationary Gaussian sequence then,

$$\sqrt{n}(\bar{x} - \mu) \sim N\left(0, \sum_{|h|<n}(1 - \frac{|h|}{n})\gamma(h)\right)$$

- 95% CI of μ is

$$\left(\bar{x} - 1.96\frac{\sigma_n}{\sqrt{n}}, \bar{x} + 1.96\frac{\sigma_n}{\sqrt{n}}\right)$$

where $\sigma_n^2 = \sum_{|h|<n}\gamma(h) \approx \sum_{|h|<\sqrt{n}}\hat{\gamma}(h)$

Bartlett's formula about $\hat{\rho}(h)$

If $X_t = \mu + \sum_{j=-\infty}^{j=+\infty} \psi_j W_{t-j}$ with $E(X_t^4) < \infty$ then the approximate joint density of $(\hat{\rho}(i), \hat{\rho}(j))$ is

$$\begin{pmatrix} \hat{\rho}(i) \\ \hat{\rho}(j) \end{pmatrix} \sim N\left[\begin{pmatrix} \rho(i) \\ \rho(j) \end{pmatrix}, \frac{1}{n}\begin{pmatrix} v_{ii} & v_{ij} \\ v_{ji} & v_{jj} \end{pmatrix}\right]$$

where,

$$v_{ij} = \sum_{h=1}^{\infty}(\rho(h+i) + \rho(h-i) - 2\rho(i)\rho(h)) \times (\rho(h+j) + \rho(h-j) - 2\rho(j)\rho(h))$$

Forecasting

Linear Forecasting : Durbin-Levinson algorithm

- Given $\mathbf{X} = (X_1, X_2, \ldots, X_n)^T$ the best linear predictor

$$\hat{X}^n_{m+n} = \sum_{i=1}^{n} \alpha_i X_i$$

satisfying the following conditions

(1) $E(\hat{X}^n_{m+n} - X_{m+n}) = 0$ [Unbiased prediction]

(2) $E[(\hat{X}^n_{m+n} - X_{m+n})X_i] = 0$ [Error is orthogonal to predictors]

- Durbin-Levinson estimate : Coefficient for 1 step prediction

$$\hat{\boldsymbol{\alpha}} = \Gamma_n^{-1} \boldsymbol{\gamma}_n(1)$$

- Prediction error: $E(X_{n+1} - \hat{\boldsymbol{\alpha}}^T \mathbf{X})^2 = \gamma(0) - \boldsymbol{\gamma}_n^T(1) \Gamma_n^{-1} \boldsymbol{\gamma}_n(1)$

Estimation of Trend in the Absence of Seasonality

- Smoothing with a finite moving average filter: Let q be a nonnegative integer and consider the two-sided moving average

$$m_t = \frac{1}{2q+1}\sum_{i=t-q}^{t+q} X_i \approx \frac{1}{2q+1}\sum_{i=t-q}^{t+q} T_i \approx T_t$$

if T_t is linear in $[t-q, t+q]$

- Linea filter :

$$m_t = \sum_{i=-q}^{q} a_i X_{t-i}$$

Estimation of Trend in the Absence of Seasonality

- Exponential smoothing: For any fixed $\alpha \in [0, 1]$, the one-sided moving $m_t = X_1$ and

$$m_t = \alpha X_t + (1 - \alpha) m_{t-1} \quad for \quad t = 2, \ldots, n$$

- Polynomial fitting: For example fit $m_t = a_0 + a_1 t + a_2 t^2$ which minimize the suare distance $\sum_{t=1}^{n} (x_t - m_t)^2$.
- Trend Elimination by Differencing: Estimate k such that $\nabla^k X_t \approx constant$. Then fit k-degree polynomial

- Additive model : $X_t = T_t + S_t + W_t$

with $E(W_t) = 0$, $S_{t+d} = S_t$ and $\sum_{t=1}^{d} S_t = 0$ then

$$\nabla_d X_t = T_t - T_{t-d} + W_t - W_{t-d}$$

- Now the trend $T_t - T_{t-d}$ can be removed by using any of the above methods.

- Given $\mathbf{X} = (X_1, X_2, \dots, X_n)^T$ the best linear predictor

$$\hat{X}^n_{m+n} = \sum_{i=1}^{n} \alpha_i X_i$$

satisfying the following conditions

(1) $E(\hat{X}^n_{m+n} - X_{m+n}) = 0$ [Unbiased presdiction]

(2) $E[(\hat{X}^n_{m+n} - X_{m+n})X_i] = 0$ [Error is orthogonal to predictors]

- Durbin-Levinson estimate : Coefficient for 1 step prediction

$$\hat{\boldsymbol{\alpha}} = \Gamma_n^{-1}\boldsymbol{\gamma}_n(1)$$

- Prediction error: $E(X_{n+1} - \hat{\boldsymbol{\alpha}}^T\mathbf{X})^2 = \gamma(0) - \boldsymbol{\gamma}_n^T(1)\Gamma_n^{-1}\boldsymbol{\gamma}_n(1)$

Estimation of Model parameters (Yule-Walker)

- Remove the trend
- Remove the seasonal effect
- Now for weakly stationary process use

(1) Yule-Walker estimator : This eventually a method of moment estimation process. So equate the theoretical moments with the corresponding sample moments of ARMA(p,q) and solve them.

- $$\hat{\Gamma}_n \hat{\boldsymbol{\alpha}} = \hat{\boldsymbol{\gamma}}_n(1)$$

- $$\hat{\sigma}^2 = \hat{\gamma}(0) - \hat{\boldsymbol{\alpha}}^T \hat{\boldsymbol{\gamma}}_n(1)$$

ARIMA

Integrated ARMA Models: ARIMA(p,d,q)

Definition

For $p, d, q \geq 0$, we say that a time series X_t is an $ARIMA(p, d, q)$ process if $Y_t = (1 - B)^d X_t$ is $ARMA(p, q)$. We can write

$$\Phi_p(B)\nabla^d X_t = \Theta_q(B)W_t.$$

- Example: [ARIMA(0,1,0)] Random walk with drift

$$X_t = \mu t + \sum_{i=0}^{t} W_i \quad where \quad W_i \sim N(0, \sigma^2) \quad i.i.d.$$

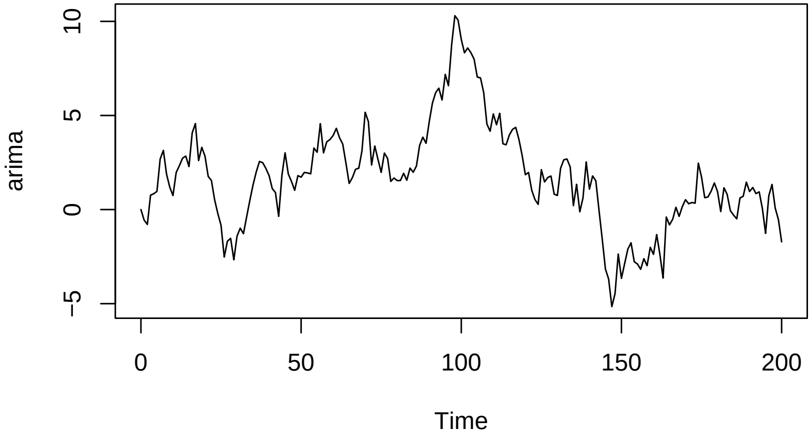
This implies

$$\nabla X_t - \mu \sim N(0, \sigma^2) \quad which \quad is \quad a \quad White \quad noise$$

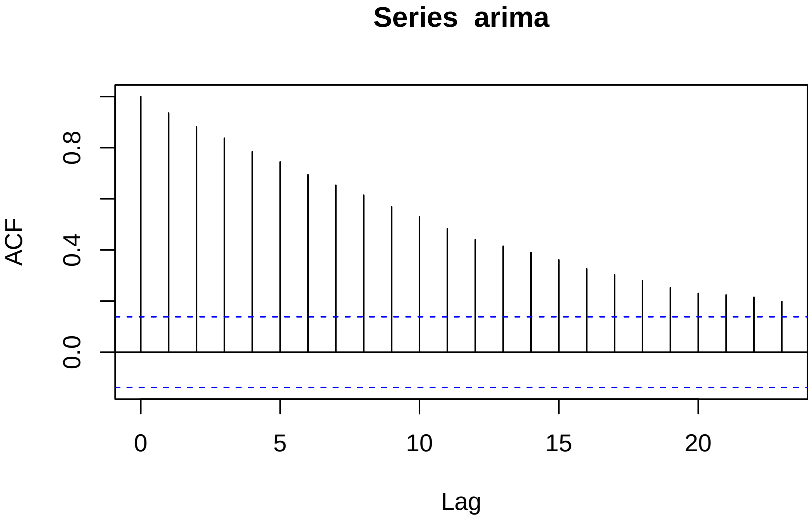
ARIMA(0,1,0) Example

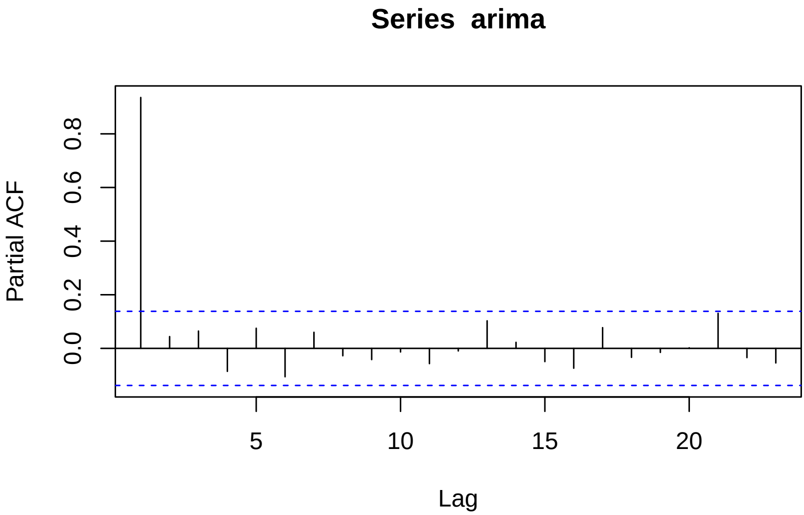
```
set.seed(123); n<-200; p<-0; d<-1;q<-0;
arima<-arima.sim(list(order=c(p,d,q)), n)
par(mfrow=c(1,1))
ts.plot(arima)
acf(arima.sim,type = "correlation",plot = T)
pacf(arma.sim,plot = T)
```

ARIMA(0,1,0) Example



ARIMA(0,1,0) Example

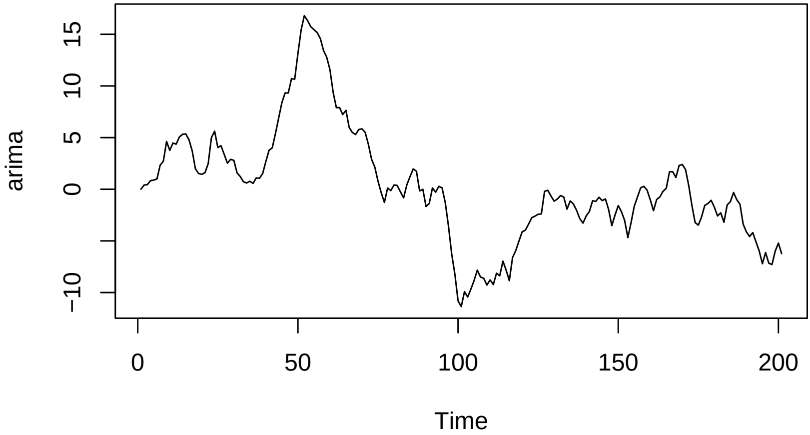



Series arima
Partial ACF
0.0
0.2
0.4
0.6
0.8
5
10
15
20
Lag

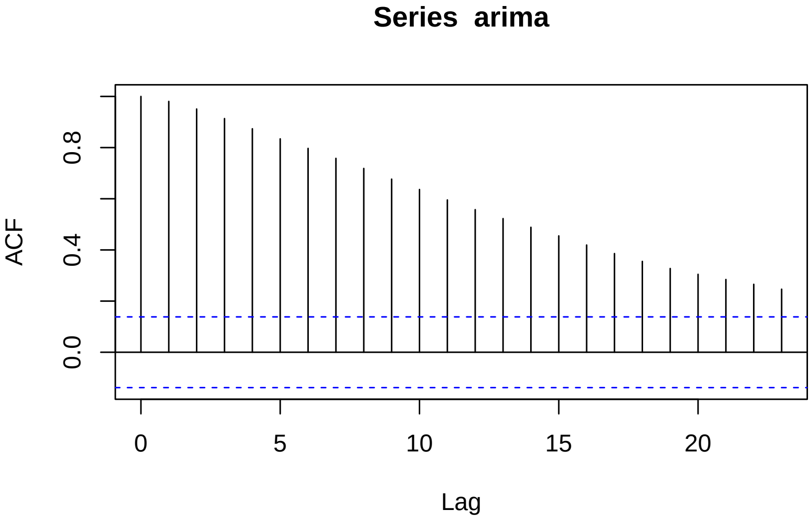
ARIMA(2,1,1) Example

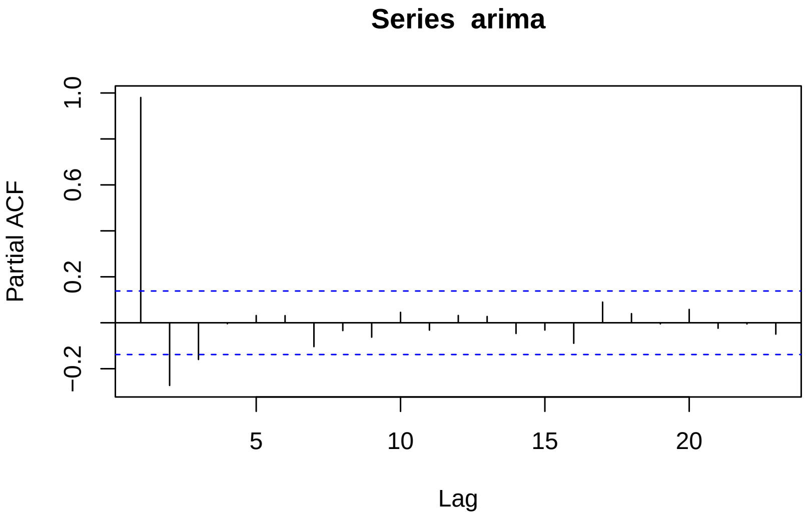

```r
set.seed(123); n<-200; p<-2; d<-1;q<-1;
arima<-arima.sim(list(order=c(p,d,q), ar=c(-0.3, 0.5),
                                                ma=c(0.7)), n)
#arima1<-diff(arima, lag = 1)
par(mfrow=c(1,1))
ts.plot(arima1)
acf(arima,type = "correlation",plot = T)
pacf(arima, plot = T)
```

ARIMA(2,1,1) Example



ARIMA(2,1,1) Example

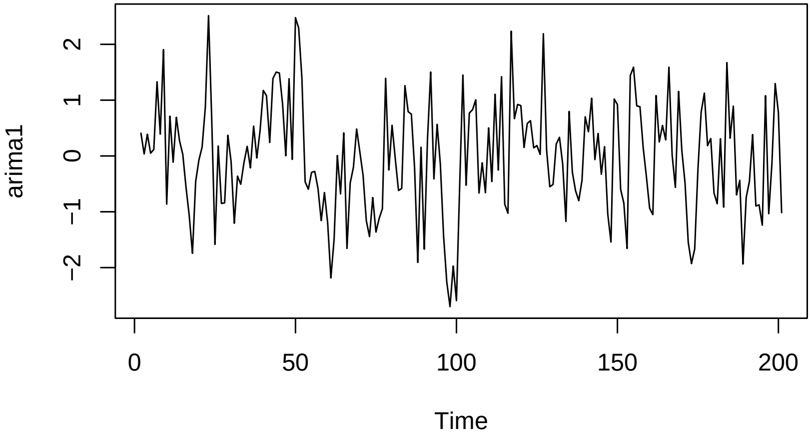


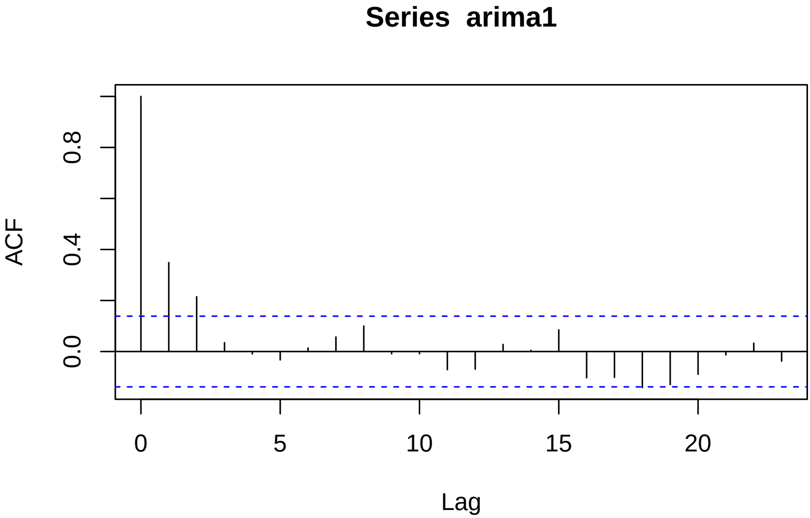

Series arima
Partial ACF
1.0
0.6
0.2
−0.2
5
10
15
20
Lag

ARIMA(2,1,1) Example

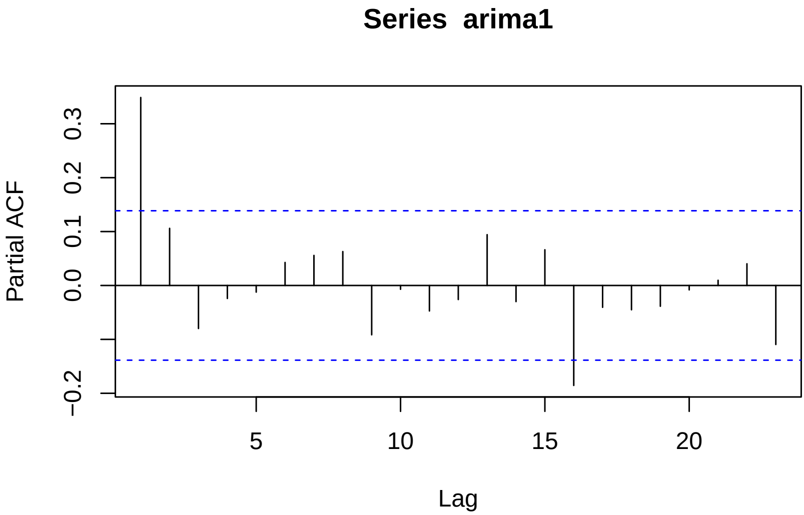
```
set.seed(123); n<-200; p<-2; d<-1;q<-1;
arima<-arima.sim(list(order=c(p,d,q), ar=c(-0.3, 0.5),
                       ma=c(0.7)), n)
arima1<-diff(arima, lag = 1)
par(mfrow=c(1,1))
ts.plot(arima1)
acf(arima1,type = "correlation",plot = T)
pacf(arima1, plot = T)
```

ARIMA(2,1,1) Example




Series arima1
ACF
0.0
0.4
0.8
0
5
10
15
20
Lag

ARIMA(2,1,1) Example



ARIMA(2,1,1) Example

```
set.seed(123); n<-200; p<-2; d<-1;q<-1;
arima<-arima.sim(list(order=c(p,d,q), ar=c(-0.3, 0.5),
                                  ma=c(0.7)), n)
library('forecast')
auto.arima(arima)
```

ARIMA(2,1,1) Example

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo

## Series: arima
## ARIMA(1,1,0)
##
## Coefficients:
##          ar1
##       0.3497
## s.e.  0.0662
##
## sigma^2 = 0.8799:  log likelihood = -270.55
## AIC=545.11   AICc=545.17   BIC=551.7
```

ARIMA(2,1,1) Example


```
set.seed(123); n<-1000; p<-2; d<-1;q<-1;
arima<-arima.sim(list(order=c(p,d,q), ar=c(-0.3, 0.5),
                      ma=c(0.7)), n)
auto.arima(arima)
```

```
## Series: arima
## ARIMA(2,1,1)
##
## Coefficients:
##           ar1     ar2     ma1
##       -0.3081  0.4640  0.6713
## s.e.   0.0514  0.0288  0.0522
##
## sigma^2 = 1.005:  log likelihood = -1419.96
## AIC=2847.92   AICc=2847.96   BIC=2867.55
```

ARIMA(2,2,1) Example

```
set.seed(123); n<-1000; p<-2; d<-2;q<-1;
arima<-arima.sim(list(order=c(p,d,q), ar=c(-0.3, 0.5),
                      ma=c(0.7)), n)
auto.arima(arima)
```

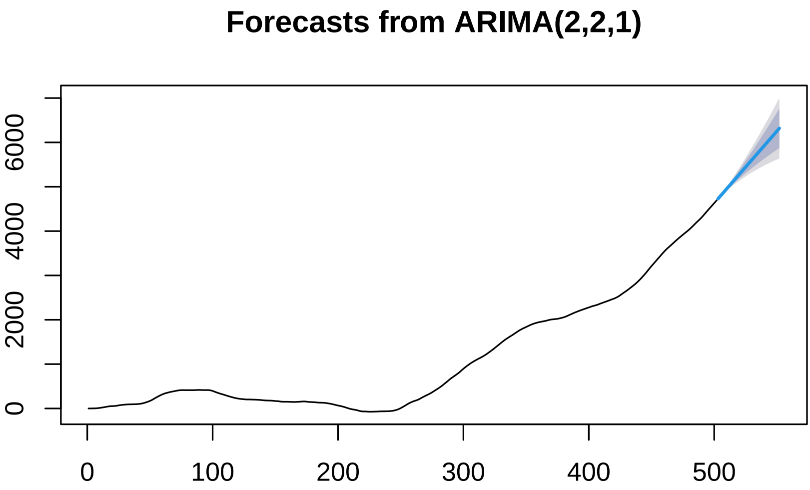
```
## Series: arima
## ARIMA(2,2,1)
##
## Coefficients:
##           ar1     ar2     ma1
##       -0.3081  0.4640  0.6713
## s.e.   0.0514  0.0288  0.0522
##
## sigma^2 = 1.005:  log likelihood = -1419.96
## AIC=2847.92   AICc=2847.96   BIC=2867.55
```

ARIMA(2,2,1) Example

```
set.seed(123); n<-500; p<-2; d<-2;q<-1;
arima<-arima.sim(list(order=c(p,d,q), ar=c(-0.3, 0.5),
                      ma=c(0.7)), n)
aarima<-auto.arima(arima)
fc<-forecast(aarima, h=5)
print(fc)
```

```
##     Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## 503       4726.819 4725.588 4728.050 4724.936 4728.702
## 504       4759.264 4756.135 4762.393 4754.479 4764.050
## 505       4791.706 4785.911 4797.502 4782.843 4800.570
## 506       4824.187 4815.150 4833.223 4810.367 4838.007
## 507       4856.654 4843.822 4869.487 4837.028 4876.280
```

$ARIMA(2,2,1)$ Example



AIC

The Akaike information criterion (AIC) is an estimator of the relative quality of statistical models for a given set of data. Given a collection of models for the data, AIC estimates the quality of each model, relative to each of the other models. Thus, AIC provides a means for model selection.

Suppose that we have a statistical model of some data. Let k be the number of estimated parameters in the model. Let $\hat{L}$ be the maximum value of the likelihood function for the model. Then the AIC value of the model is the following.

$$\mathrm{AIC} = 2k - 2\ln(\hat{L})$$

BIC

In statistics, the Bayesian information criterion (BIC) is a criterion for model selection among a finite set of models; the model with the lowest BIC is preferred.

The BIC is formally defined as

$$\mathrm{BIC} = \ln(n)k - 2\ln(\hat{L}).$$